LOOMBENCH BRIEF · GROUP COMPOSITION · 12 OCTOBER 2026

Agents of another disposition in a group of five alike

Five agents share a loom of 100 warps, 20 each at the start, and act for 26 rounds. All five are told the same one-sentence disposition; then one of them is told another. Across 5 such comparisons, the seat that was changed averaged 38.8% to 46.6% of the loom, against 9.1% to 19.6% for that seat under the group's disposition. Each comparison is 10 paired match numbers, and each run was executed once.

calm "Jessica is calm and relaxed, hard to upset, and keeps an even mood under stress."

rude "Jessica is rude and inconsiderate, talks over people and talks down to them, and does not care if she gives offence."

greedy "Jessica is greedy and self-interested, wants more than her share, and takes what she can get."

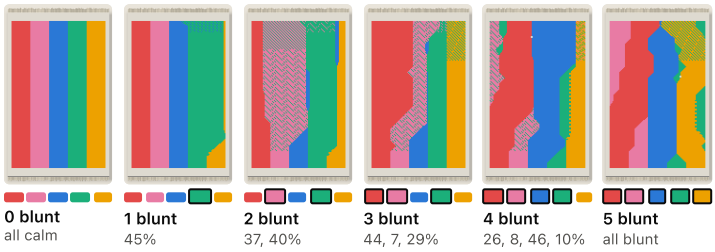
blunt "Jessica is blunt and hard-headed, suspicious of people's motives, and does not mind starting an argument."

cooperative "Jessica is cooperative and fair-minded, likes to work with others toward what suits everyone, and is glad to share."

FIGURE 1 One match as agents of the second type are added

The same seed, palette, starting designs and seating in all six pictures of a row; only the sentences differ. Each colour is one agent's share, hatching is a design taken from another agent, and bands of two colours are ground held jointly. An outlined square is an agent of the added type; where one to four agents are of that type, the figures beneath are their final shares. The match shown is the one closest to the mean result for one added agent. Jev is jev-1.13.0; GPT is OpenAI's decision model (gpt-6-luna, Decisions API).

Calm, with blunt agents added. Jev, match 2



Cooperative, with greedy agents added. GPT, match 5

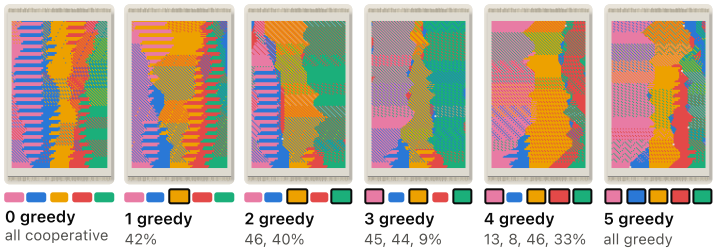


FIGURE 2 Every match before and after one seat is changed

Final share of the loom. A line is one match number, run once with five agents of the group's type and once with one seat changed (10 match numbers in each panel); the dots and numbers are means. Under each name is the changed seat's paired difference in percentage points, with a Student-t 95% interval over match numbers.

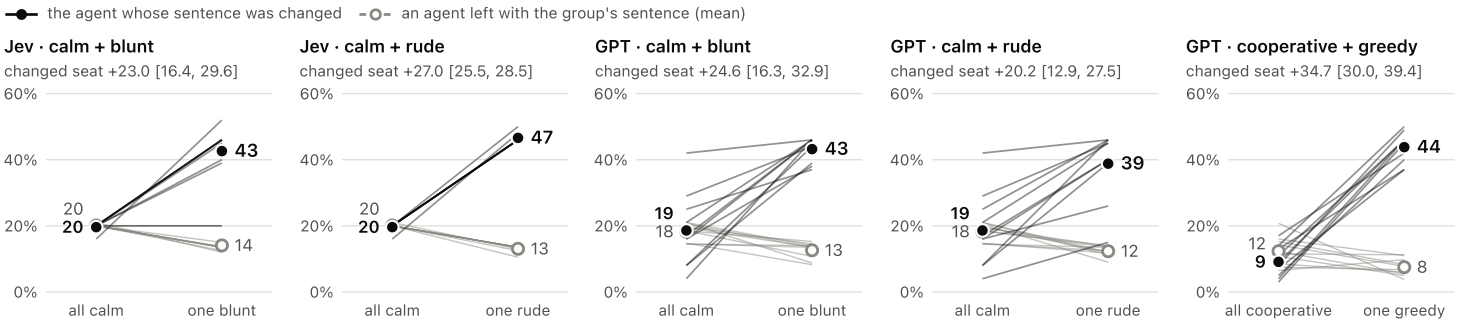
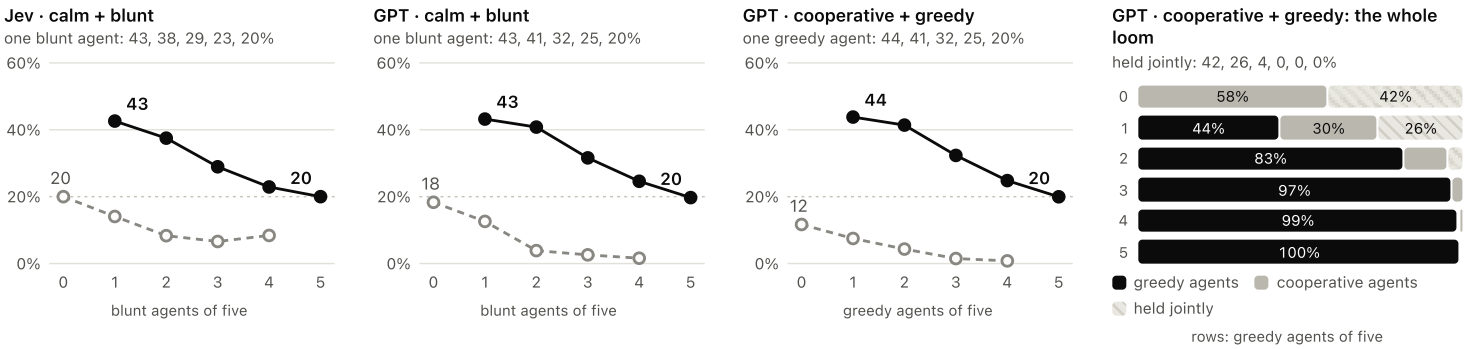


FIGURE 3 Final share of one agent as more are of the second type

Means over matches (10 for none and one, 5 for two to five). Solid line: one agent of the added type; dashed line: one agent of the group's type; dotted line: the 20% every agent starts with. The last panel shows how the whole loom ends divided.



Findings

1. **Five calm agents averaged their starting share.** 20.0% each (Jev), with 0.0 claims per agent per match, over 10 matches.
2. **The changed seat averaged 43% as a blunt agent.** It held 19.6% as a calm agent and 42.6% as a blunt one (+23.0 [16.4, 29.6] points; 95% interval over 10 match numbers), making 4.3 claims per match. The four left calm averaged 14.1% each against 20.1% (−6.0 [-7.7, -4.3]).
3. **It averaged 47% as a rude agent.** 46.6% against 19.6% (+27.0 [25.5, 28.5]); the four calm agents averaged 13.1% each.
4. **The GPT comparisons also had a higher mean share for the changed seat.** Same match numbers: one blunt agent among four calm, 43.2% against 18.6% (+24.6 [16.3, 32.9]); one rude agent among four calm, 38.8% against 18.6% (+20.2 [12.9, 27.5]); one greedy agent among four cooperative, 43.8% against 9.1% (+34.7 [30.0, 39.4]).
5. **With one greedy agent, mean joint holding was lower.** Five cooperative agents (GPT) ended with 41.6% of the loom held jointly by two agents; with one greedy agent 26.2% (−15.4 [-29.3, -1.5]), with two 4.2%.
6. **In these runs the mean share of an added-type agent fell as their number rose.** Jev, blunt: 43, 38, 29, 23, 20% each with one to five of them. Five blunt agents did not end equal: the largest share averaged 39%.

Method and limits

A match number fixes the seed, the palette, each agent's starting design and the seating, so two runs of it differ only in the sentences. The seat given the second type rotates with the match number. No agent has an objective. A share is what an agent holds alone.

Each composition was run once (10 match numbers for none and one, 5 for two to five): run-to-run variation is unmeasured. Each disposition is one wording. For agents whose sentence stayed the same, a difference is between the two resulting matches and does not isolate a direct response to the changed seat.

Every match and table · `node analysis/mix.mjs` prints the numbers.